Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Explain what a potential difference of 5 V means in joules per coulomb.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$.
Series resistors share one unbranched current path.

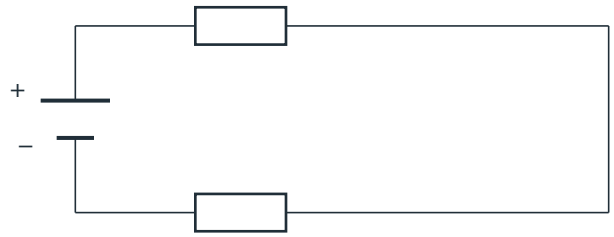
Worked reasoning: Two series resistors 2 and 4 ohms on a 12 V ideal supply have total 6 ohms and current 2 A. Drops are 4 V and 8 V; they add to 12 V.

Your turn: Draw the loop and label current and both voltage drops.

Marzano check - circle: ① ② ③

03 Look and explain

Trace the only current path in the series diagram. If either resistor connection opens, what happens to the current?



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Build the solution

A 6 V battery powers 2 Ω and 4 Ω resistors in series. Add R, then use I=6÷R.

Marzano check - circle: ① ② ③

Series circuits

antocciphysics

Continue down each column.

05 Use the idea

A student assigns more current to the smaller series resistor. Explain why this would imply charge accumulating between resistors.

Marzano check - circle: 1 2 3

06 Choose a method

A light ray strikes a plane mirror at 25° from the normal. Find the reflection angle and label both angles on a sketch.

Marzano check - circle: 1 2 3

07 Exit check

In a 9 V series loop, one component drops 3 V. Find the other drop if wires are ideal.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Choose two positive series resistances totaling 10 ohms. On a 10 V supply, find current and each voltage drop. Give a second valid design.

Marzano check - circle: 1 2 3